



Participant Handbook

Sector
Green Jobs

Sub Sector
Waste Management

Occupation
Aggregator

Reference Id
SGJ/Q6201, Version 2.0; NSQF Level 3



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Agri – Residue Aggregator

This book is sponsored by

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Shri Narendra Modi
Prime Minister of India

“

Skilling is building a better India.
If we have to move India towards
development then Skill Development
should be our mission.

”



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About this Book

Galvanizing Organic Bio Agro Resources Scheme was launched in February 2018 to increase cleanliness and development in the village by converting the waste and fertilizer received from agriculture and animal husbandry to biogas.

Similarly, the Ministry of Petroleum and Oil companies started the Patel called Sustainable in September 2018, which aims to provide continuous option for affordable transport, Sustainable Alternative Turds Affordable Transportation which will be beneficial for both vehicle users as well as farmers and entrepreneurs.

Under these schemes, Skill Council for Green Jobs has developed the job roll of agriculture and remaining aggregator with aimed to manage the waste from agriculture and animal husbandry and to meet its supply and demand. This participant book has been designed for training of agriculture and remaining aggregator according to the qualification pack SGJ/Q6201, which is available on qualification pack <http://sscgj.in/publications/qualification.pack/>.

This book is designed keeping in mind the minimum educational qualification of participants. This book is in language of the picture with minimum literature to read the participants easily. This book will provide participants the necessary knowledge and skills to do their work correctly.

The Skill Council for Green Jobs is grateful to the team of the 'Kontent Edge' whose participant has a valuable contribution in the development and content writing of the booklet.

The signs used in the book are listed below.

Symbols Used



Key Learning
Outcomes



Steps



Role Play



Tips



Notes



Unit
Objectives

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Employability Skills

It is recommended that all trainings include the appropriate Employability skills Module.

Content for the same can be accessed:



[Employability skills workbook](#)

Annexure

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1. Introduction

Unit 1.1 – Farm waste and its types

Unit 1.2 – Role of Agricultural Residue
Aggregator

Chapter Highlights

After reading this chapter, you will be able to:

1. Identify farm waste and its sources
2. Identify the types of farm waste
3. Identifying the market for farm waste
4. Know the benefits of managing agricultural residues
5. To know the importance of work of Agricultural Residue Aggregator
6. Step By Step Biomass Supply Chain Management
7. Identify the work and responsibilities of an Agricultural Residue Aggregator
8. To know the tools and equipment used by the Agricultural Residue Aggregator

Unit 1.1: Farm waste and its types

Unit objectives

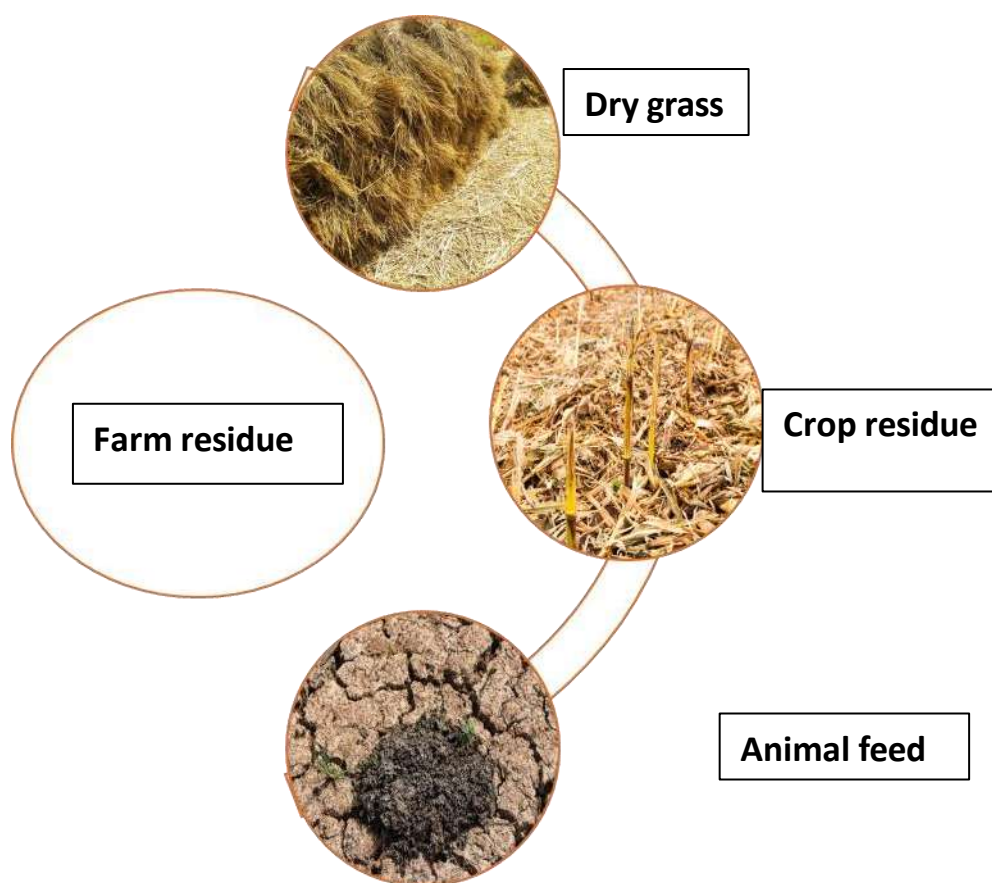


After going through this unit you will be able to do the following aspects

1. identify farm waste and its sources
2. identify the types of farm waste
3. Identifying the market for farm waste
4. Know the benefits of managing agricultural residues

1.1.1 Farm Waste

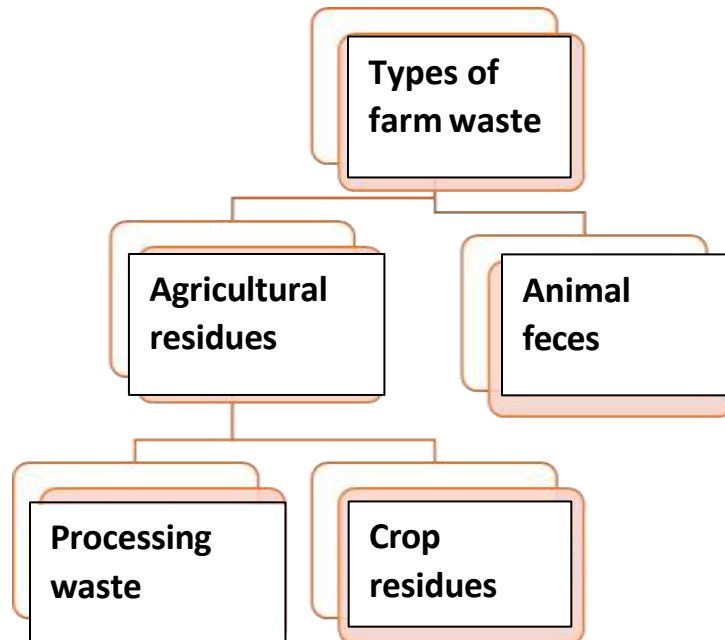
Garbage and leftover material from farming and agriculture work such as dry grass, straw, tree, remaining part of plant, animal material, fertilizer chemical etc are called farm waste. Some examples of farm waste are shown in the picture below:



Pic.1.1.1: Examples of farm waste

1.1.2 Types of Farm waste

Materials like animals, machines, seeds and fertilizers are used in farming and agriculture. After the use of all these, many types of farm waste and garbage settle. The main types of farm waste are shown in the picture below.



Pic.1.1.2: Types of farm waste

Agricultural waste from crops

Waste from crops is also called agricultural residue, in which rice, wheat, maize, jowar, which comes after harvesting crops like millet, etc., some examples of farm waste are shown in the picture below:



Residue



Leaves &
Stem



Dried Husk, Bark,
Oil Cake



Stalk

Pic.1.1.3: Types of farm waste

Waste agricultural residue from crops can be divided into two parts on the basis of harvesting and processing of crops. The picture below shows both the parts of crop residue:

Primary waste



Agricultural residues produced in the field at the time of harvest.

Paddy straw, sugarcane tops, Maize stalk, empty coconut shell and palm leaves etc.

Secondary waste



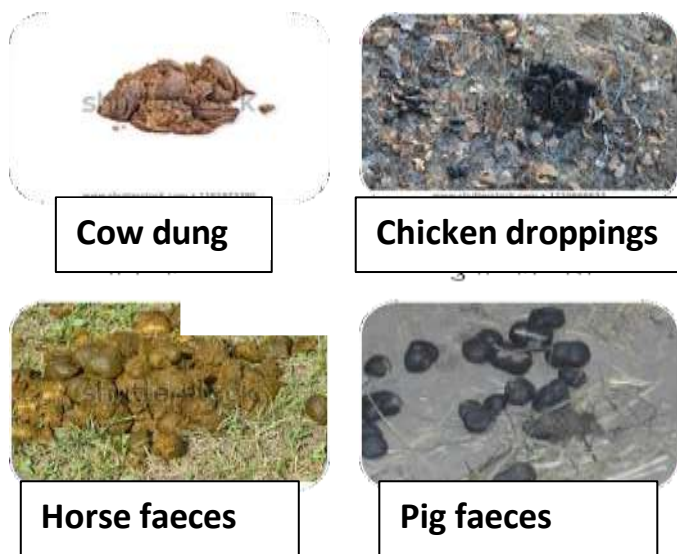
Agricultural residues from processing.

Paddy husk, bagasse, maize kernel, coconut husk, coconut coir sawdust, sawdust, palm

Pic.1.1.4: parts of agricultural residues

Waste from Animal Husbandry

Livestock waste means cow, buffalo and poultry excreta which is used to make biogas, it is also a part of agriculture waste, the waste of different animals is shown in the picture below:



Pic.1.1.5: Picture of animal feces

Apart from making biogas, cow excrement can also be used as a fuel. Cow dung has been used as fuel in open hearths since traditional times but in view of the increasing pollution in the environment and health problems admitted in women due to this, the central government of India has launched an awareness campaign not to use these stoves.

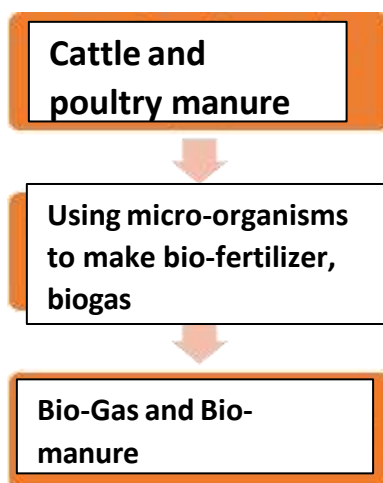
In the picture below, cow dung is shown to be used for making bio-fertilizer.



Picture 1.1.6: Use of cow dung

Microorganisms are required to make biogas from animal waste. This bacteria helps in rotting manure and converting it into biogas.

The same process is explained below :



Picture 1.1.7 Biogas production from cow dung



Pic.1.1.8 biogas production from cow dung

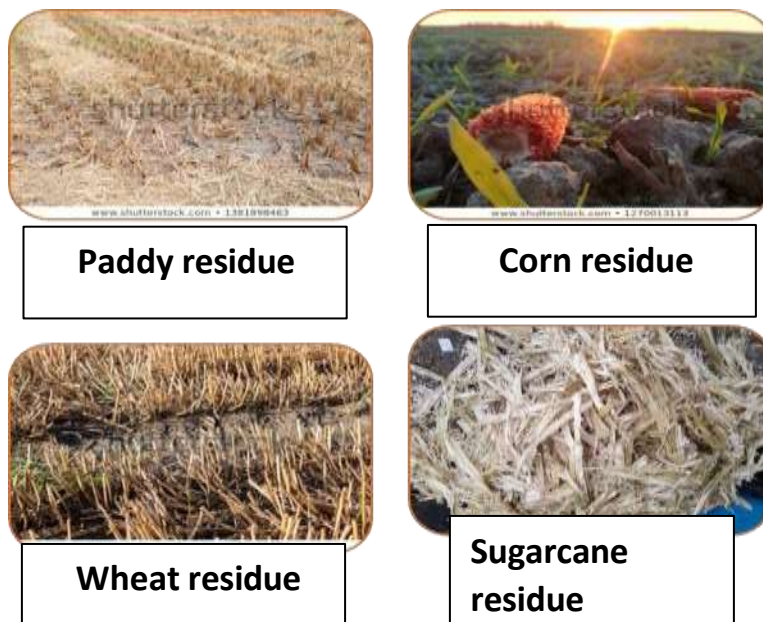
1.1.2 Sources of Agricultural Residue

Paddy, wheat and maize are the major food crops in India, which contribute a major portion of the total farm waste production. According to a report, about 600 million tonnes farm waste is generated in India every year. About 200 to 240 million tonnes of the total farm waste generated in India is surplus i.e. surplus waste. This waste includes all types of waste from different crops. The major sources of agricultural residues are mentioned below.

The major sources of agricultural residues are mentioned below:

Crop	Residue	Major Producer
Paddy	straw and chaff	West Bengal, Uttar Pradesh, Andhra Pradesh
wheat	straw	Uttar Pradesh, Punjab and Haryana
Maize	Cattle feed Gulli and silk	Karnataka, Maharashtra, Rajasthan, Uttar Pradesh, Andhra Pradesh, Bihar
sugarcane	Bagasse	Maharashtra, Uttar Pradesh

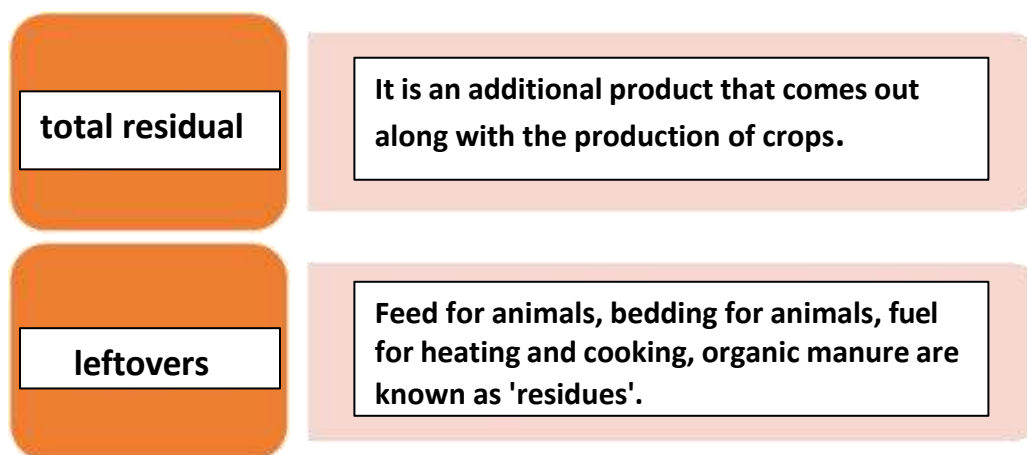
Wastes of major crops are shown in the picture below:



Pic.1.1.9 Waste of major crops

1.1.3 Market for agricultural residues

On the basis of quantity of agricultural residues, the market of agricultural residues is divided into two parts. These two parts are explained below:



Pic.1.1.10: Total and left over residue

The remaining residue can be used for organic manure production. Cereals contribute the most to the remaining residue all over India. Other crops that give a large amount of residual are the most important oilseeds, pulses and sugarcane. This farm waste can also be used to make pallets. This vest can be put to good use by using new and emerging technology. Biogas made from this waste can be used to generate energy and in vehicles.

1.1.4 Benefits of managing agricultural residues

Managing agricultural waste means using it properly or giving it to the collector so that it can be used properly or dispose of it. Managing agricultural residues is very important and has the following advantages:



Protecting the environment



Helpful in making bio compost



Production of bio-gas



Electricity production



Use in transport



Use in cooking

Pic.1.1.11: Benefits of managing agricultural residues

Exercise



1. Write the name and type of agricultural residue given below:













2- Write the advantages of agricultural residue management in the given table

1.	
2.	
3.	
4.	
5.	

Unit 1.2: Role of Agricultural Residue Aggregator

Unit objectives



After going through this unit you will be able to:

1. To know the importance of the work of agricultural residue aggregator
2. Knowing the steps of biomass supply chain management
3. Identify the work and responsibilities of an agricultural residue aggregator.
4. To know the tools and equipment used by agricultural residue aggregator

1.2.1 Importance of work of Agricultural Residue aggregator

India is the second largest agricultural economy in the world, due to which a large amount of farm waste is generated here. Due to not knowing the method of disposal of this waste properly, most of the farmers burn it in the open.

Burning of waste in this way has become a big challenge not only for the environment but also for human health. For this reason, the activity of burning the waste has now been banned by the government and a provision has also been made to fine the farmers who do so. This prohibited activity i.e. burning of crop waste is shown in the picture below.



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Pic.1.2.1: Open burning of agricultural residues

To properly dispose of this waste, a biomass supply chain management has been developed, in which the agricultural residue aggregator plays an important role. In biomass supply chain management, agricultural waste aggregator buys agricultural waste from farmers and sends it to biomass processing centers so that it can be used properly.

1.2.2 Biomass supply chain management

In biomass supply chain management, different types of waste related to agricultural sector such as farming, harvesting, processing, storage and transportation are collected and used to make bio fuel.

All the steps of biomass supply chain management are explained in the picture below:

To collect

- Agricultural waste is collected in this step.

Sorting out

- The waste that can be used to make bio-fuel is sorted.

To transport

- After sorting the waste is sent to the processing center.

To store

- Waste is stored in the processing center until it is used.

Pic.1.2.2: Steps of Biomass Supply Chain Management

1.2.3 Duties & Responsibilities of Agricultural Residue Aggregator

Agriculture remains cricketer has to do many things which are following:

Calculating market demand and supply for agricultural residues taking into account seasonal movements etc.

Liaison between supplying farmers and buyers

Creating a Collection Point for Agricultural Residue

Creating a Collection Point for Agricultural Residue

Sale and Marketing of Agricultural Residue

Calculating market demand and supply for agricultural residues considering weather, climate etc

Liaison between supplier farmers and buyers

Creating a Collection Point for Agricultural Residue

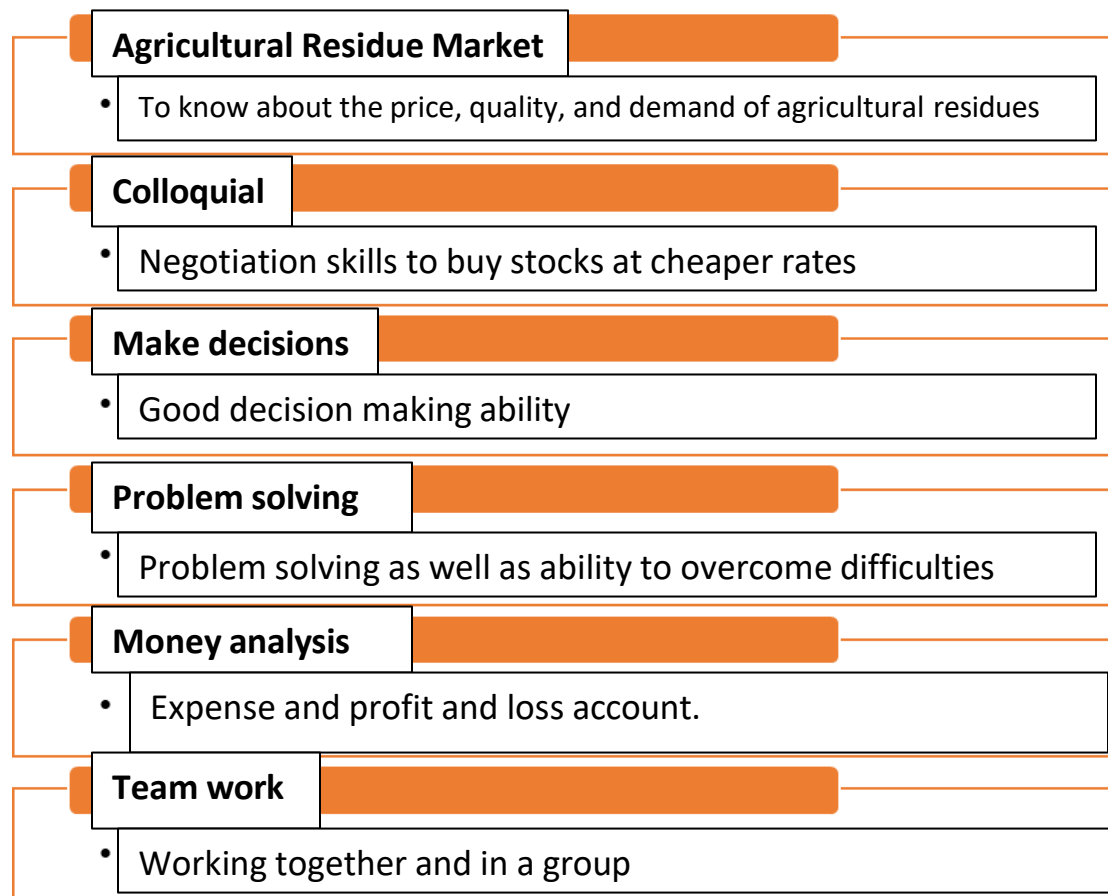
Thickening and Stacking of Agricultural Residue

Sale and Marketing of Agricultural Residue

Pic. 1.2.3: Duties & Responsibilities of Agricultural Residue Aggregator

1.2.4 Efficiency for Agricultural Residue Aggregator

To perform all the functions of Agricultural Residue Aggregator efficiently a person should have the following skills



Pic.1.2.4: Skill set of Agricultural Residue Aggregator

1.2.5 Tools and Equipments used by Agricultural Residue Aggregator

Baler machine

- to pick up and press agricultural residues into thicker bales,
- to maintain quality
- To ease handling storage and transportation



Pic.1-2-5: Baler machine

Tractor:

- To connect and run the dealer machine
- To attach a loader or other equipment to a bell-pump if necessary.
- To move gutters from one place to another for storage etc.



Pic.1.2.6: A Tractor

Bale stacker

- To attach to the tractor
- To line the bale
- To assemble them properly



Pic.1.2.7: A bale stacker

Exercise



- 1- Which of the following is not one of the functions and responsibilities of an Agricultural Residue Aggregator?
 - a. Linkage between farmers and buyers
 - b. collection of agricultural residues
 - c. Sale of Agricultural Residue
 - d. Cutting and making gutters of agricultural residues in the field
 - e. compaction and stacking of agricultural residues
- 2- Name the tools and equipment of the agricultural residue aggregator given below:



Scan the QR code or click on the link to watch the related videos



<https://www.youtube.com/watch?v=Gpo2E-0dDXU>



<https://www.youtube.com/watch?v=2YLPB52zVX8>

Scan the QR code or click on the link to watch the related videos



<https://www.youtube.com/watch?v=um00IUEX65s>



<https://www.youtube.com/watch?v=WBdVkWxPPFQ>



2. Assessing Demand and Supply of Agricultural Residue



Unit 2.1 – Assessing Demand and Supply of Agricultural Residue

Unit 2.2 – Identifying suppliers for planning supply of agricultural residue stock



Chapter Highlights

After reading this chapter, you will be able to understand the following

Points:

1. Know the current state of the agricultural residue market
2. To know the trend of agricultural residues in the market
3. Identifying and contacting supplier/farmers
4. Negotiate with the buyer
5. Fixing price and placing order
6. Understanding the steps to maintain stock
7. Determining the Right Quality of Agricultural Residue

Unit 2.1: Assessing Demand and Supply of Agricultural Residue

Unit objectives

After reading this unit, you will be able to understand the following points:

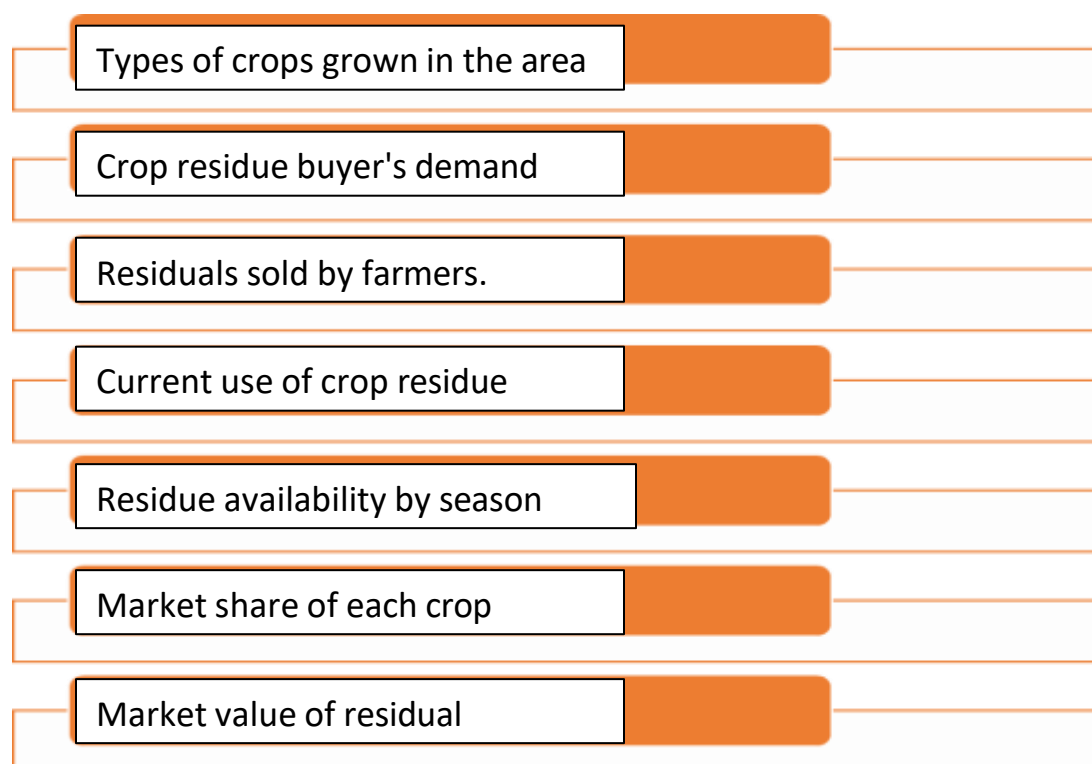
1. Know the current state of the agricultural residue market
2. To know the trend of agricultural residues in the market

2.1.1 Understanding the Current State of the Agricultural Residue

Market

An agricultural residue aggregator can use several methods to locate an agricultural residue market. He can understand the status of the agricultural residue market by taking into account the trend of the last several years in the area. Points like last year's cultivation and residue yield, demand for last agricultural residue of monsoon, etc. are important.

The Market Of Agricultural Residue Depends On Many Points. The key points to understand the state of the agricultural residue market are shown in the diagram below:

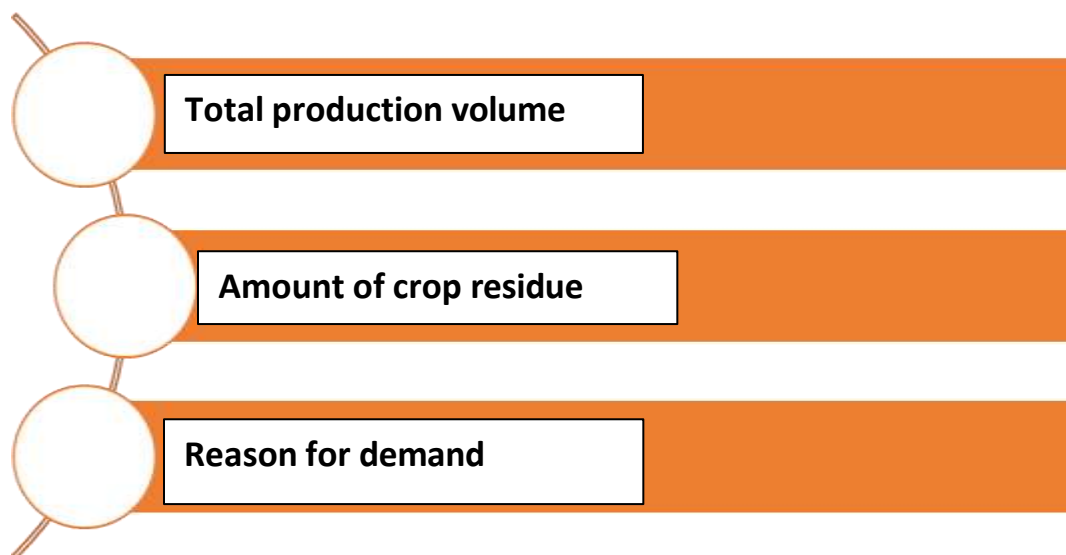


Pic.2.1.1: Key Points of the Market Position

2.1.2 Understanding the Phases of the Agricultural Residue Market

After understanding the status of agricultural residues, the aggregator should understand the market trends. With this, he can identify his buyers and sellers present in the market and plan his work according to their demand and supply.

The diagram below shows some points to understand the trend of the market:



Pic.2.1.2: key market trends

Exercise



1. Mention the requirement of agricultural residues in the market ?
 1. _____
 2. _____
 3. _____

2. Give the correct sequence of contacting agricultural residue suppliers.
 1. Keep the supply constant
 2. Listing of buyer's requirement
 3. Connect with the farmer
 4. Ensuring supply meets demand

Unit 2.2: Identifying suppliers for planning supply of agricultural residue stock Buyers & Sellers

Unit objectives

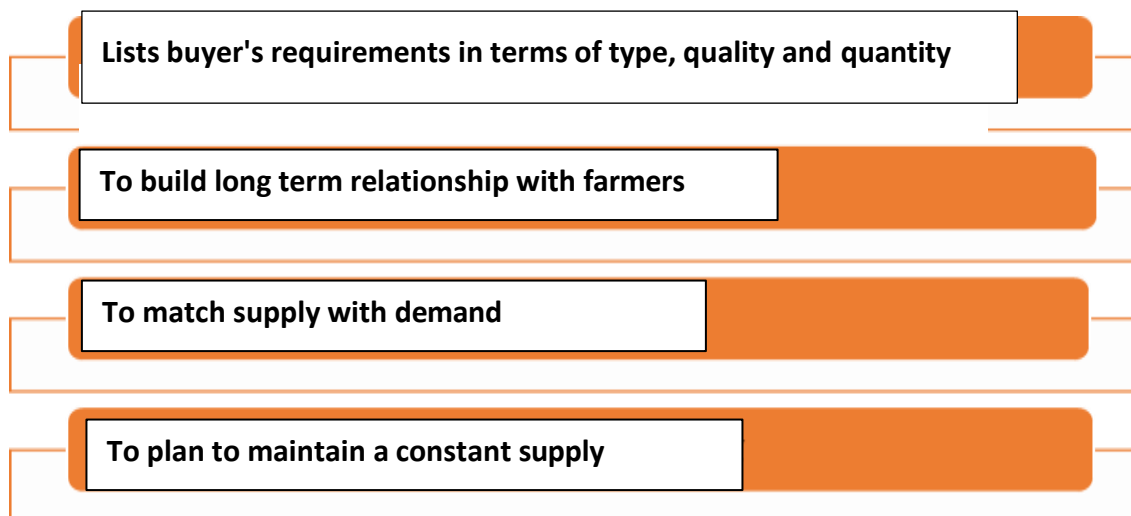
After reading this unit, you will be able to understand the following points:

1. Identifying and contacting suppliers / farmers
2. Negotiating with the buyer
3. Fixing price and placing order
4. Know the steps to maintain stock
5. Finding the Right Quality of Agricultural Residue

2.2. Planning to contact suppliers / farmers

After calculating the entire agricultural residue, the aggregator has to contact the farmers to buy the residue. Contacting the farmers means that the aggregator should understand the demands of different buyers and maintain good relations with the farmers so that both of them can further increase their business in the future.

Reasons for needing to contact the Agricultural Residue Aggregator are shown in the diagram below:

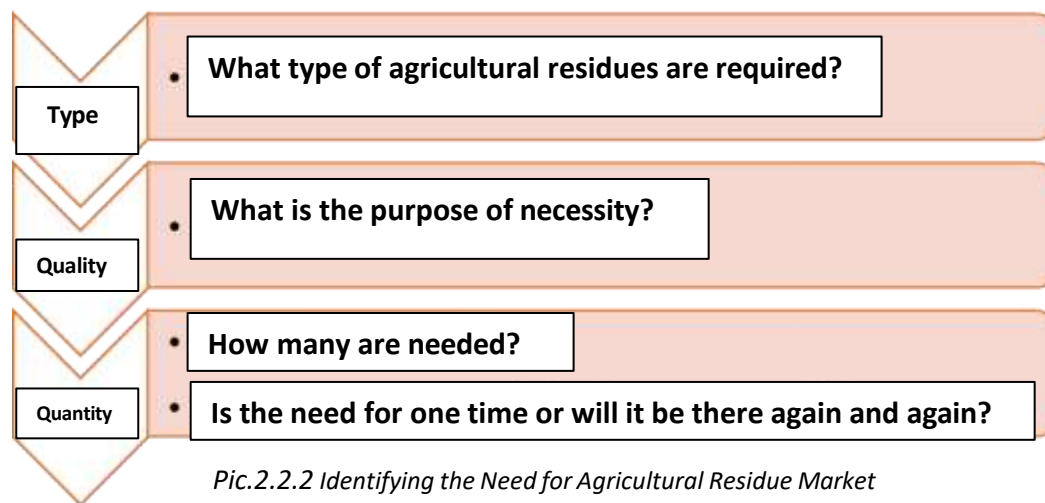


Pic.2.2.1 Contact farmers

Agricultural residue collectors must coordinate with the farmers and ensure that the supplies are of the quality demanded. In order to maintain a constant supply, the agro-residue collector should interact with the farmer and also with the buyers from time to time.

2.2.2 Contact buyers

The agriculture residue aggregator should be in constant touch with the buyer to understand the requirement of the agricultural residue. The aggregator should coordinate with the buyer on the following points.



Pic.2.2.2 Identifying the Need for Agricultural Residue Market

The following figure shows the purpose of agricultural residue requirement in the market:



For animal feed



To cover rural houses



For industrial use



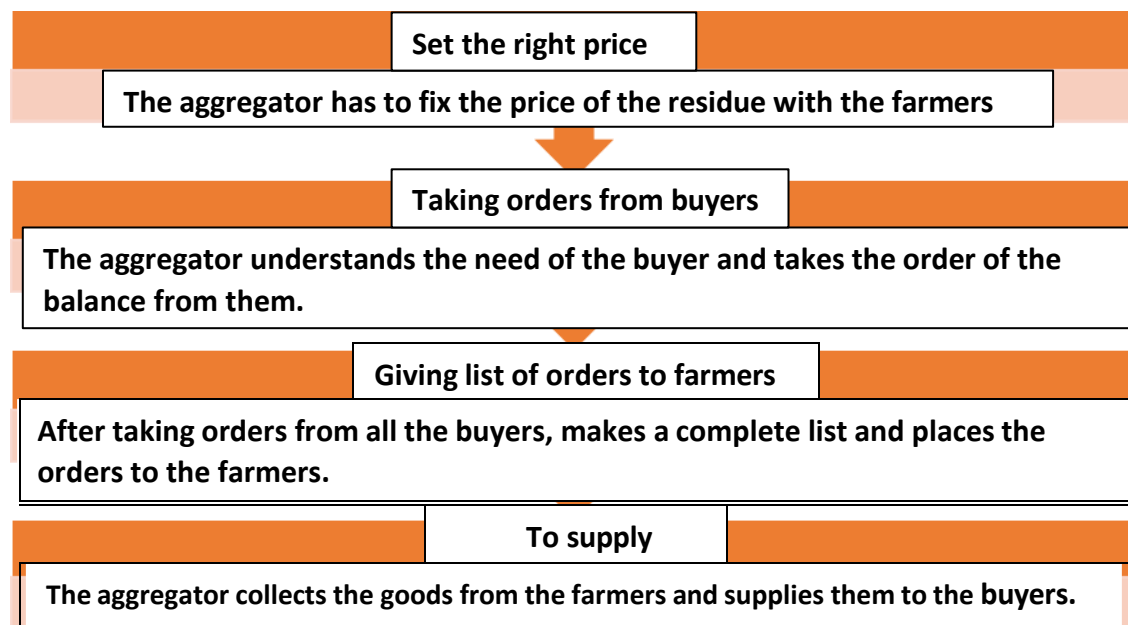
For making organic fertilizers



To make fuel for domestic and industrial use

2.2.3 Price fixing and ordering

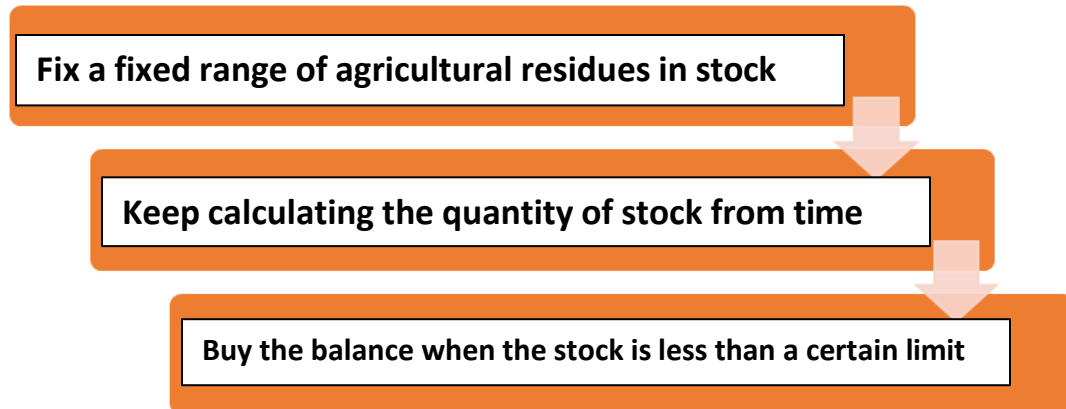
In order to buy agricultural residues, the aggregator has to maintain constant contact with the farmers and also take care of the quality of the residues. The steps required to buy the residues are shown in the picture below:



Pic.2.2.4: Steps to Buy Agricultural Residue

2.2.4 Maintain Stock

An aggregator has to maintain some additional stock keeping in mind its buyers so that it can be met in time when additional or uncertain demand arises. The steps required to maintain regular stock are explained in the diagram below:



Pic.2.2.5: Regular stock availability

2.2.5 Agricultural Residue Quality Check

Maintaining the quality of agricultural residue is one of the major functions of the aggregator. The quality of agricultural residue can be maintained by the following methods:

1. **Density** . While checking the quality of agricultural residue, the aggregator should keep in mind that the density of the bundle of residue should neither be too high nor too low. Low density makes it very difficult to handle the stock and in high case this can cause problems with transportation.

The figure below shows bundles of residues of the wrong and correct densities:



Wrong



Correct

Pic.2.2.6: Methods of checking quality standards

2. **Moisture content** . The bundle of agricultural residue should not have more moisture than required because if the moisture content is high, then this bundle will not be used for a long time and can also damage other bundles. The picture below shows bundles of remnants of true and very high moisture content:



Wrong



Correct

Pic.2.2.7: Methods of checking quality standards

Exercise

1. Identify the correct sequence of approaching farmers for agricultural residue stock
 - a. Ordering
 - b. price setting
 - c. Complete account of supply with respect to Residue quantity, time of harvest, quality etc.
 - d. To obtain an estimate of the requirement in terms of type, quality and quantity
 - e. Coordinating with both farmers and buyers to ensure a continuous supply
- 2- Identify which of the following is not one of the quality aspects of stocking agricultural residues.
 - a. ash content
 - b. moisture content
 - c. thickness
 - d. physical properties
 - e. fix price

Exercise related

1. Forecast the Agriculture Residue Kiva demand and supply for a given stock so that planning can be done in advance. This forecast will give complete knowledge about the future of the market and decide the right price to buy and sell the stock.

Devices:

- calculator
- paper and pen
- Phone

2. Demonstrate the steps involved in contacting farmers and identify the standards set for agricultural residues

Devices:

- paper and pen
- Phone



3. Purchase of Agricultural Residue Stock from Nodal Point



Unit 3.1 - Establish nodal point for collection

Unit 3.2 - Collecting and receiving agricultural
residue stock



Chapter Highlights

After reading this chapter, you will be able to understand the following points:

1. Identifying the steps to prove collection point for purchase of stock from farmers.
2. Coordinate with suppliers to buy stock at low prices.
3. Identifying the tools and equipment to be used at the collection point.
4. Explaining the entire process of collecting/receiving agricultural residues.
5. Managing Purchased Stock

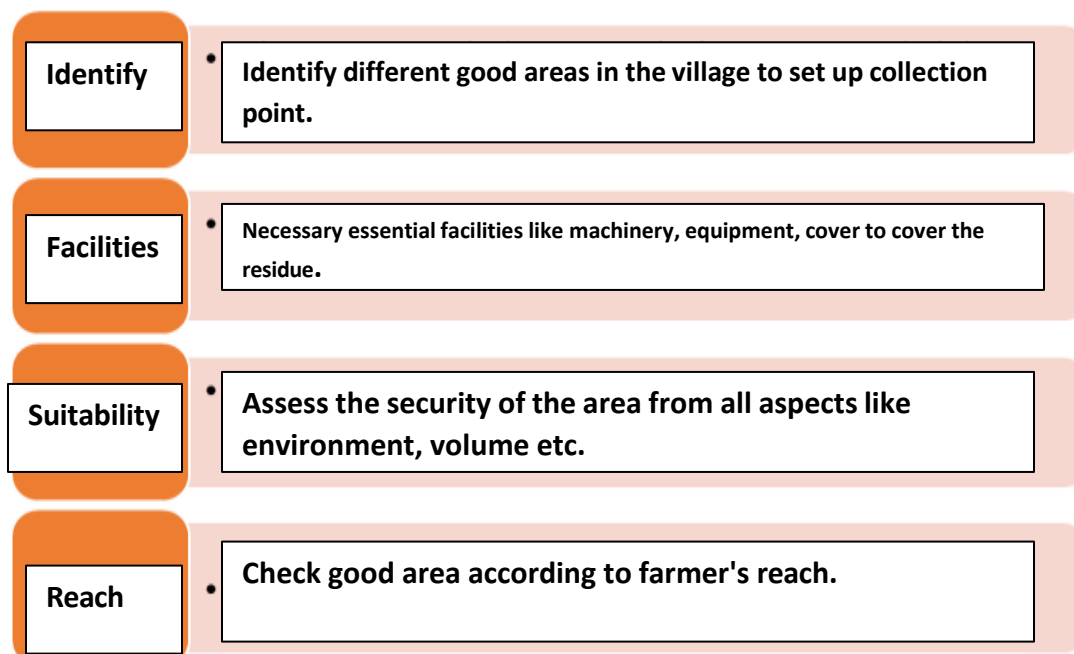
Unit 3.1: Setting up collection points for agricultural residue collection

After reading this unit, you will be able to understand

1. Identifying area to prove collection point
2. Indicate the steps for proper installation of the collection point.

3.1.1 Identification of area for setting up collection point

The collection point is the area where the agricultural residue is brought from the fields by the farmers. The agricultural residue aggregator is designed to collect the agricultural residue keeping in mind the convenience of the farmers as well as exposure to weather i.e. sun, rain etc. A suitable area should be identified for this. Depending on the quantity of agricultural residue, there may be one or more collection points located in the center of each village. The selection of a good collection point depends on the following factors:



Pic.3.1.2: Location and utility of the collection point

3.1.2 Setting up a collection point

Once the suitable area for the collection point is identified, the Agriculture Residue Aggregator should take steps to set up the collection point to collect the stock from the farmers of that particular area.

Such collection points are shown in the figure below:



Pic.3.1.3: Agricultural Residue Collection Point

- Set up nodal points for collection of residues in different villages for the sitting areas of the farmers supplying the village specific.
- Ensure the availability of the necessary tools and equipment to manage agricultural residues delivered by suppliers.
- Calculate the number of laborers required according to the estimated quantity of agricultural residues to be collected at these nodal points.

3.1.3 Informing farmers about collection points

After identifying the area and establishing the nodal point for the collection of agricultural residue from different farmers in different villages, the agricultural residue aggregator should inform the farmers about the same.

The picture below shows farmers collecting the residue:

- Ensure the farmers about the nodal point related to the collection of their agricultural residues where they can collect their stock.
- Inform them about the likely schedule of collection at these nodal points to manage the stock.

3.1.4 Contacting farmers for purchase of agricultural residues

Once the collection points are set up in different villages, the agricultural residue aggregator should contact the farmers.

- Meet with the farmers according to their harvesting time to check the availability of agricultural residues considering the type, season, climate etc.
- Estimate the possible quantity and negotiate for the best price.
- Order agriculture and balance to reach the designated collection point.
- Schedule the delivery of agricultural residues from the suppliers / farmers. Scheduling makes it easy to manage the stock.



Pic.3.1.4: Agricultural Residue Suppliers

3.1.5 Providing tools and equipment at the collection point

After placing the order with the supplier/farmer, the aggregator of agricultural residue should ensure the availability of necessary tools and equipment at the designated nodal point for collection.

Below are the important tools used in the job :



Scales

- To weigh bales of agricultural residue.



Claw

- For bulk collection of cut material.



Baler

- To give round or rectangular shape to the collected residue.



Forklift

- To lift and carry a bundle.



Tractor

- For safe transportation of stock.

Pic.3-1-5: Tools and Equipment for Agricultural Residue Management

Note: If cutting machine is needed, include raker. Raker is an agricultural implement attached to the tractor, which is used for cutting straw, stalks, etc.

A raker machine is shown in the figure below:



Pic.3.1.6 Raker

Exercise

- 1- Identify the correct step involved in setting up a collection point for agricultural residue.
 - a. Ensuring the availability of tools and equipment
 - b. Checking the Quality of Stock
 - c. Estimating the decided price
 - d. Weighing Agricultural Residue Stocka
- 2- Identify the thousands and the following sets of equipment needed for agricultural residue collection.



a. _____



b. _____



c. _____



d. _____



e. _____

Unit 3.2 Collecting and buying agricultural

Unit objectives:



After reading this unit, you will be able to understand the following points:

1. Identifying the process of collecting agricultural residues in terms of quality, quantity and cost.
2. Listing the Steps for Buying Agricultural Residue
3. Transport and stock management of agricultural residues.
4. Coordinating with suppliers for procurement of agricultural residues.
5. Identifying the tools and equipment to be provided at the collection point

3.2.1 Agricultural waste collection

The Agricultural Residue Aggregator should check the thickness and other essential properties of the Agricultural Residue collected at the nodal point with respect to quality and quantity before fixing the correct price for the Agricultural Residue.

All the steps for collecting agricultural residues are explained below with pictures:

1- **Moisture test** :Is the residue dry or wet?



wrong



correct

2- **Thickness check** : Is the thickness and moisture content of the residue low or high?



Wrong



Correct

3- **Weighing Agricultural Residue** : Measure the quantity with the help of scales.

The picture below shows some farmers weighing agricultural residues:



- 4- **Quality checks** : Is the quality of biomass as per the demanded quality?

Checking the quality of an aggregate residue is shown in the figure below:



- 5- **Cost estimation** : Taking care of quality quantity time and place.
The diagram below shows an aggregator estimating expenses:



Pic.3.2.1: Stages of collecting agricultural residues

3.2.2 Purchase and accounting of agricultural residues

Once the price is negotiated and fixed, the agricultural residue aggregator should purchase the selected stock from the supplier/farmer and make payment for the same, which should be properly recorded in the records.

Below are all the steps of this process with pictures:



Measure quantity and quality



Write full account



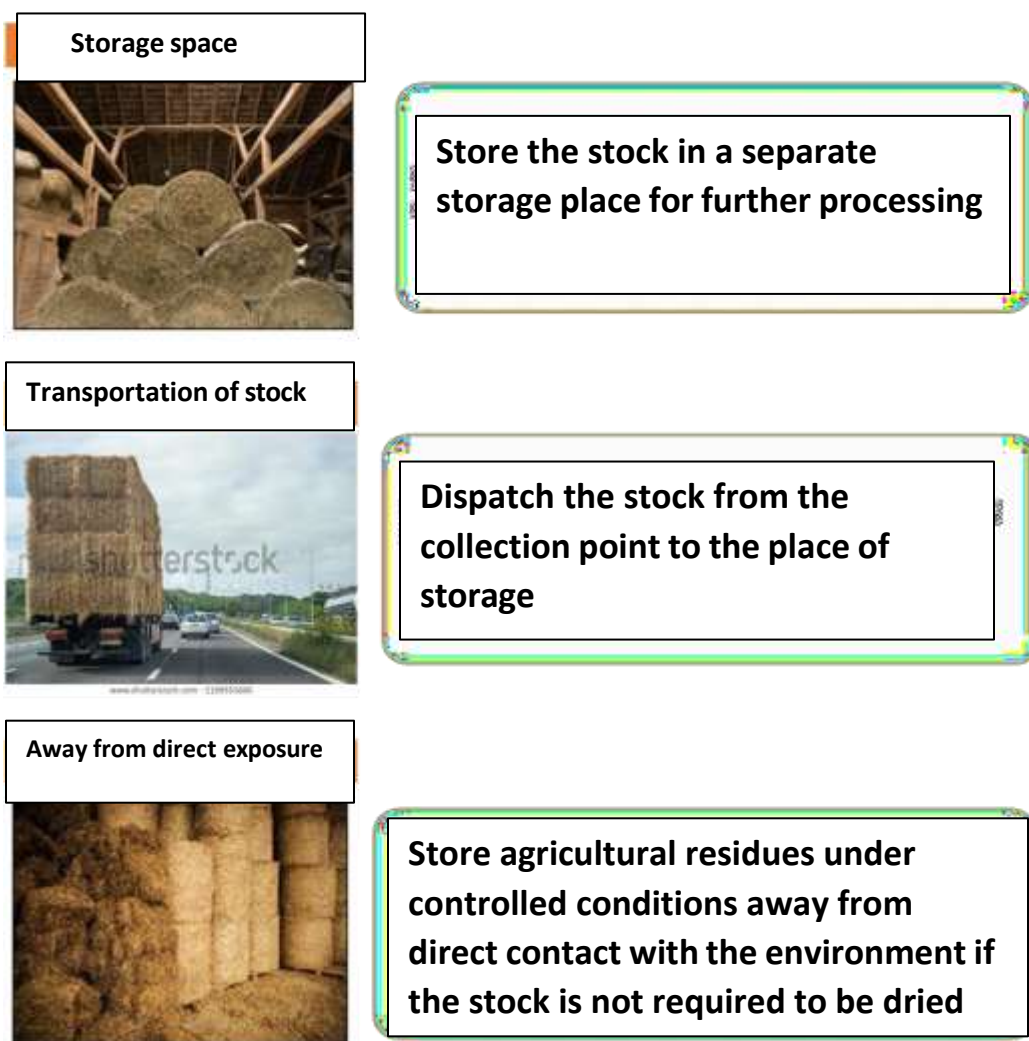
Buy and Pay

Pic.3.2.2: Purchase assessment

3.2.3 Transporting and stocking agricultural residues

To maintain the quality and quantity of the stock of agricultural residues procured and collected, they should be transported to the place of storage quickly. Stock should be stored optimally for easy handling and transportation.

The diagram below shows some of the essentials for optimal placement and transportation of stock



Pic.3.2.3: Important things for storage of agricultural residues

Exercise



- 1- Identify the correct sequence of steps involved in the purchase of agriculture and land.
 - a. stock weighing
 - b. stock quality check
 - c. price estimation
 - d. checking thickness and moisture

- 2- Identify the correct method of storage and transportation of agricultural residues.
 - a. Transport by any means and stock it anywhere
 - b. Transport through trucks and stock it in storage space
 - c. transport the stock without pressing it and stock it loosely
 - d. Store in any way in this storage location



Exercise related

- 1- Demonstrate setting up of a nodal collection point for agricultural residues procured from farmers and suppliers in different villages.

Make sure it is centrally located and farmers should not have any difficulty in reaching it.

Device:

- Balance
- Claw
- baler
- Tractor
- forklift

- 2- Demonstrate the collection of agricultural residues at the nodal point by coordinating with the farmers. Check the quality, quantity and condition to fix the price once the agricultural residues are properly delivered. Complete the purchase by making payments and accounting.

Devices:

- Laptop /Pen /Computer
- Official format/document for recording data



4. Packing and Storing Compacted Agricultural Residues



Unit 4.1 - Compaction and packing of
agricultural residues

Unit 4.2 - Storing agricultural residues

SGJ/N6203

Chapter Highlights



After reading this chapter, you will be able to understand the following points:

1. To grade the agricultural residue and tell the method of sorting.
2. Identifying the characteristics of agricultural residues prior to baling.
3. Explain the process of making bales of agricultural residues and collecting the bales.
4. Listing the steps for packing a bundle
5. Identifying bale storage precautions and methods.

Unit 4.1: Compaction and packing of agricultural residues

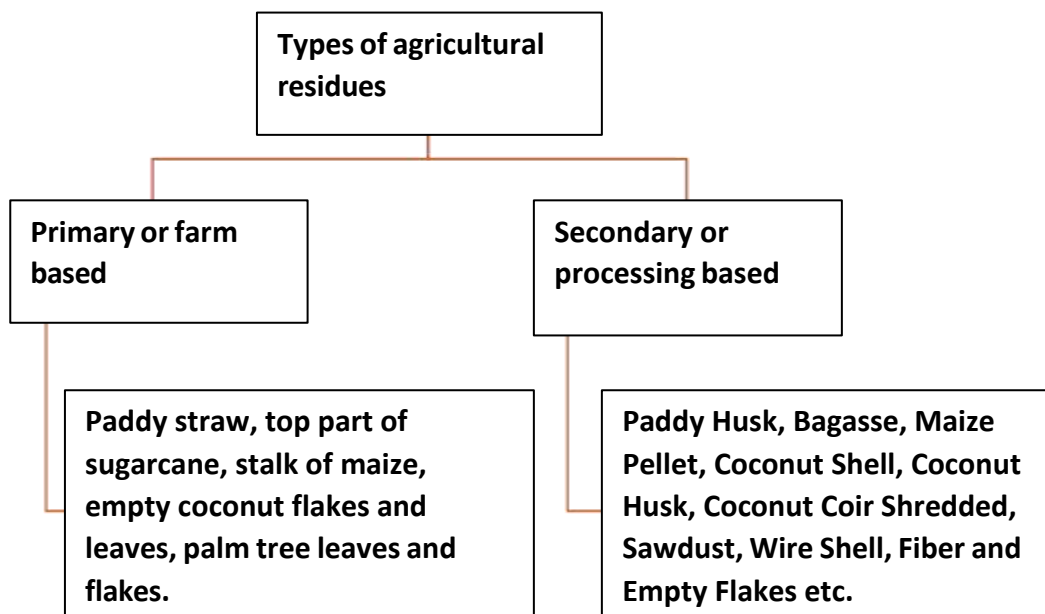
Unit objectives

After reading this unit, you should be able to:

1. Grading of agricultural residues and methods of sorting.
2. Identifying the characteristics of agricultural residues prior to baling.
3. To know the process of making bales of agricultural residues.
4. Determining Bale Quality

4.1.1 Grading and sorting of agricultural residues

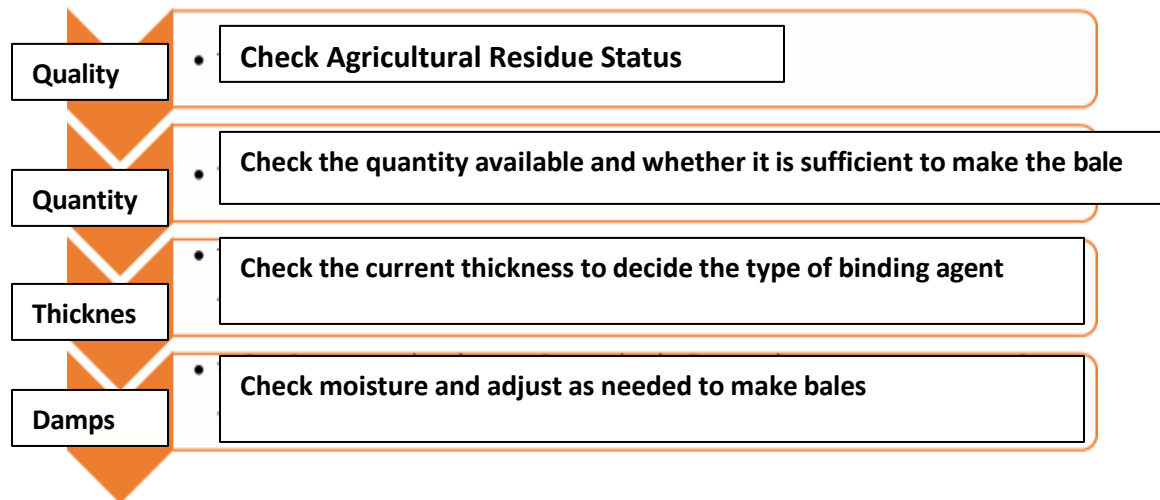
Agricultural residues collected from supplier farmers must be graded and sorted to avoid the presence of heat-generating residues during baling. Check the entire bale and a list of types of agricultural residues. Segregate the collected agricultural residues according to their type, nature and condition.



Pic.4.1.1: Determining the grade and sorting of agricultural residues

4.1.2 Checking Agricultural Residue before Baling

Following are the important factors to check agricultural residue:



Pic.4.1.2: Agricultural Residue Testing

- Due to the high moisture content of the residue, irregular shape and less thickness required, it is very difficult to handle, transport, store and utilize agricultural residue in its original form, therefore baling is a good solution.
- Baling is also an effective process to overcome the constraints of typical biomass utilization such as uneven particle size and cost of transportation.
- After baling, the energy-generating capacity of biomass increases without changing its chemical composition.

4.1.3 Baler operation

Baling is one of the most traditional forms of adding or mixing agri-residue which is done with the help of a baler that compresses the material. Bales can be square, rectangular or round depending on the type of baler is used.



Circular dealer machine



Rectangular dealer machine

Pic.4.1.3 Different shapes of baler

The pressure plays a very important role in the formation of bundles. Due to low pressure, the thickness decreases whereas with increasing pressure, the thickness of bundles increases significantly, so the pressure should be improved.

The picture below shows all the steps to operate the baler.

			
Before starting the process of making agro. residue, check the baler.	Put the agriculture. residue in the baler properly.	Maintain the pressure of Bell's piston press..	Operate the machine properly.

Pic.4.1.3: Baler Machine Operating Steps

4.1.4 Collecting bales

Baling through baler machine produces thicker bales as compared to the earlier raw residue which makes it easier to handle and transport. Once the thick bales are prepared, they are properly collected from the baling machine. To make room for the next bale, collect all the bales neatly in one place.

The bundles assembled are shown in the picture below:



Pic.4.1.4 Stacking of Bales

Exercise



Arrange the grading and sorting steps of agricultural residues in the correct sequence.

- a. Review collection
- b. Segregate Agricultural Residue

Unit 4.2: Storing agricultural residues

Unit objectives



After reading this unit, you should be able to:

- Tell the steps to pack the bales
- Identify the precautions and methods of storing the bale.

4.2.1 Packing the bale

Once the compacted bales are collected with a baler machine, they must be properly packed for storage and transportation. Packaging consists of wrapping several plies of thin material to form a strong package that can be tied with rope or twine. Net wrap is even used to protect them from weathering. The packaging prevents them from expanding back into their former shape from their new shape.

A tied bundle is shown in the figure below:



Pic.4.2.1 Packaged Bales

4.2.2 Equipment needed for packing bales

Baler tying machine can be used for tying the bale with rope. The main purpose of tying the bale is to keep the whole material together without scattering so that there is no problem in transportation later.

The figure below shows bundles tied with rope:



Pic.4.2.2 Bales binding sheet and sheet forming machine

4.2.3 Ways to store bales

Agricultural residue **bales** can be tied up and stored in many different ways depending on region, weather, etc.

Bales of agricultural residues need protection from the weather, and are best stored indoors.

Ways to store bales in order of high cost to low cost include:

- An enclosed building with a concrete floor
- An open building made of concrete floor reusable tarpaulin on concrete
- Storage on concrete outside
- Outside ground storage
- Underground

4.2.4 Store the bundles

1. Bales available under roof should be stored in the right place.
2. Stack the bales and line them up correctly.
3. Maintain proper temperature moisture and controlled environment during storage.

The bundles collected at one place are shown in the picture below:



Pic.4.2.2 To store the bundle

4.2.5 Precautions to be taken while storing the bundles

- Bundles of agricultural residues should be stored in such a way that they do not get wet.
 - Bundles should be stacked in such a way that the moisture content is minimum
- Bundles should never be exposed to heat or any potential source of ignition.

Exercise

- 1- Identify the correct method of packing the bundles.
 - a. using wrapping machine
 - b. using twine
 - c. use thread
 - d. using net wrap
- 2- Identify the correct methods of storing the bundle.
 - a. stack the bundle
 - b. put them in order
 - c. store at the right temperature
 - d. Maintain a moisture and controlled environment during storage

Exercise related



- 1- Demonstrate how to make a bundle using a machine to make storage and transportation easier. Check the quality, quantity, name and thickness of the bundle before making it, so that it is fit for purpose.

Device:

- Baler machine

- 2- Perform proper packing of bundles to maintain quality and store in proper storage location. Maintain proper temperature, humidity and controlled environment during storage.

Devices:

- storage space
- packing material
- wrapping machine if needed

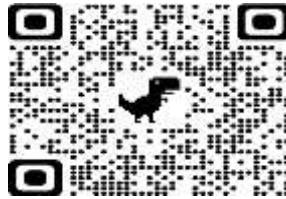
Scan the QR code or click on the link to watch the related video



<https://www.youtube.com/watch?v=yIkYjJsWhek>



<https://www.youtube.com/watch?v=DSP0xk7Lpls&t=120s>



<https://www.youtube.com/watch?v=75Rh1Q4Sk6A>

Baling and packing of agricultural residues



5. Sales and transportation of agricultural residues



Unit 5.1 - Marketing and Selling of Agricultural Residue

Unit 5.2 - Transporting agricultural residue



Chapter Highlights



After reading this chapter, you will be able to understand the following points:

1. Understand the selling position of the agricultural residue market.
2. Identifying and coordinating with potential buyers for Agricultural Residue Waste.
3. Explaining how to create a sales order
4. Maintaining Customer Relations
5. Identifying the Correct Method of Transporting Agricultural Residues
6. Choosing the Right Transportation

Unit 5.1: Marketing and Sales of Agricultural Residue

Unit objectives



After reading this unit, you will be able to understand the following points:

1. Understanding the Selling Position of the Agricultural Residue Market
2. Identifying Potential Buyers for Agricultural Residue Waste
3. Coordinate with them for Agricultural Residue Waste.
4. Explains how to create a residual sales order.
5. Maintaining relationships with customers

5-1-1 Understanding the Selling Position of the Agricultural Residue Market

Agriculture residue aggregator should take quality and quantity into consideration to understand agriculture residue sales trend. Collect agriculture residue sales trend data of previous crop as well as current crop. Data calculation for preparation of sales and marketing strategy for the collected stock of agricultural residues from farmers/suppliers.

5-1-2 Identifying Potential Buyers for Agricultural Residue Waste

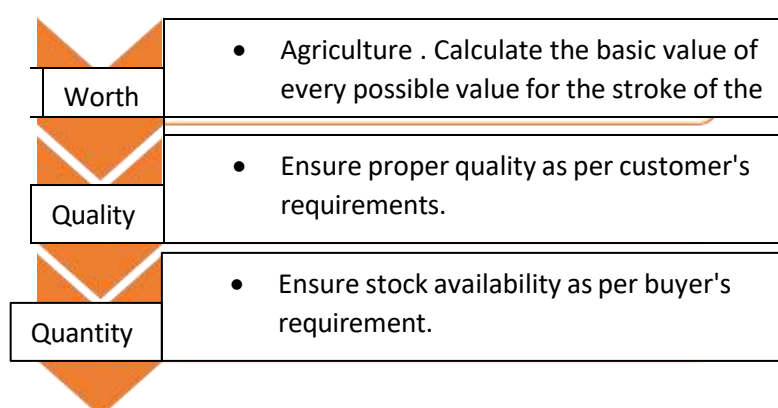
List the type, quality, quantity and specific type, if any, of the agricultural residue collected. The agricultural residue aggregator should identify potential buyers based on an estimate of the agricultural residue sales trends in the current market. Potential buyers may include industries to business operations. Those that produce paper, electricity, ethanol, gasoline, diesel, etc. depending on the agricultural residue, potential buyers include those who have bought stock before or are new to the business.

5-1-3 Contacting potential buyers for Agricultural Residue Waste

Contact potential buyers to know their current demand by type, quality and quantity. Let them know about current stock availability of agro residue. Choose the appropriate buyer in a given market situation in which supply meets demand in all respects.

5-1-4 Sales order creation

The Agricultural Residue Aggregator should keep in touch with the buyer selected to sell the stock of Agricultural Residue to create a Sales Order and finalize everything related to the residues including price, quality, quantity etc.



Pic.5.1.1: Sales order creation

5.1.5 Maintain customer relationship

The agricultural residue aggregator should fulfill its responsibility of maintaining the customer relationship in the following manner.

Maintained regular touch with buyers /customers of Agri-residue stock from order placement till delivery

Take care of buyers/customers queries regarding the stock ordered.

Contact the buyers/customers from time to time to know and meet any incoming demand.

Facilitate buyers/customers as and when required.

Pic.5.1.2: Ways to maintain customer relationship

Exercise



1- Know the right way to create sales orders&

correct	wrong
Bargain the price	Fixed price
Quality as per customer's requirement	Production quality
Quantity as required	Quantity as per availability

2- Mention the steps involved in maintaining a customer relationship-

- a. _____
- b. _____
- c. _____
- d. _____

Unit 5.2 Transporting agricultural residues

Purpose of the unit



After reading this unit, you should be able to:

1. Identifying the Right Way to Transport Agricultural Residues
2. Identifying the most appropriate vehicle for transportation
3. Explaining the measures to be taken while transporting.

5-2-1 Methods of Transporting Agricultural Residue

Transportation is mainly concerned with loading and unloading operations and conveying the biomass to the delivery point.

Below are three common modes of transportation based on distance shown with diagrams:

1. An animal cart
2. Tractor trolley (for short distance)
3. Truck (for long distance)



Pic.5.2.1: Different modes of transportation of agricultural residues

1. Selection of tractor.trolley according to the cost of transportation such as fuel consumption per hour, cost of fuel, hourly wages of transport driver, distance of transport, carrying capacity of transport and speed of transport is done
2. Freight costs for long distance transportation by truck are taken into account.

5-2-2 Selection of transport

The agricultural residue aggregator should decide on the appropriate mode of transportation based on the cost, quality, time constraints, safety, etc. from the sending location to the receiving location.

The decision should be taken after careful consideration of the following points:



Pic.5.2.2: Transportation of agricultural residues

- The most important factors affecting the cost of transporting biomass are its physical form and quality.
- Bales of greater thickness will carry more material over less per unit distance. The smaller the thickness of the straw, the more expensive it is to transport.
- Truck transport is widely well developed and is generally the cheapest mode of transport but becomes expensive as the distance traveled increases.
- Check the most suitable route for on.time delivery, estimate the cost and then decide the best mode of transport based on the conditions of the area.

Exercise

1- State the correct method of transportation of agricultural residues for the following situations .

- a. When the delivery point is in another state:
- b. When delivery point is within Raj but Puri is more:
- c. When the delivery point is at a short distance: 2.

Write the correct measures to be taken while transporting agricultural residues.

Correct	Wrong
load to capacity	overload
proper packaging	no packaging
save stock	no security measures
Timely delivery	untimely delivery

Experimental

Experiment 1

Demonstrate Marketing and Selling of Agricultural Residue in the market by selecting the right buyer, arriving at the right price, ensuring the quality standards as per the requirement of the customers.

Devices:

1. Past trend and current record of sale of Agricultural Residue
2. paper/pen
3. list of potential buyers
4. Phone

Symbolic solution:

- Collect and analyze sales trend data.
- Contact potential buyers to select a buyer for the current stock.
- Negotiate the price considering the quality, quantity and type of agricultural residue stock.
- Create Sales Order

Experiment 2

Demonstrate proper transportation of agricultural residue stock through the most suitable mode of transport from the sending point to the delivery point based on location, cost, quality, time constraints, safety, etc.

Devices :

- mode of transport
- Stock of Agricultural Residue to be delivered
- workers to load stock
- paper/pen
- Phone

Indicative Solution:

1. Calculate the cost of delivery by different modes of transport.
2. Choose the best mode of transport for agricultural residue delivery
3. Load the stock securely and set it properly.
4. Save the stock from falling or any possible damage.
5. Pack the stock from all sides and tie it tightly.
6. Co-ordinate with the driver to ensure timely delivery of stock.
7. Confirm delivery arrival from buyer.
8. Choose the best mode of transport for agricultural residue delivery
9. Load the stock securely and set it properly.
10. Save the stock from falling or any possible damage.
11. Pack the stock from all sides and tie it tightly.
12. Co-ordinate with the driver to ensure timely delivery of stock.
13. Confirm delivery arrival from buyer.



6. Maintaining basic health and workplace safety



Unit 6.1 - Maintaining health and safety at work



SGJ/N6205

Unit 6.1: Maintaining health and safety at work

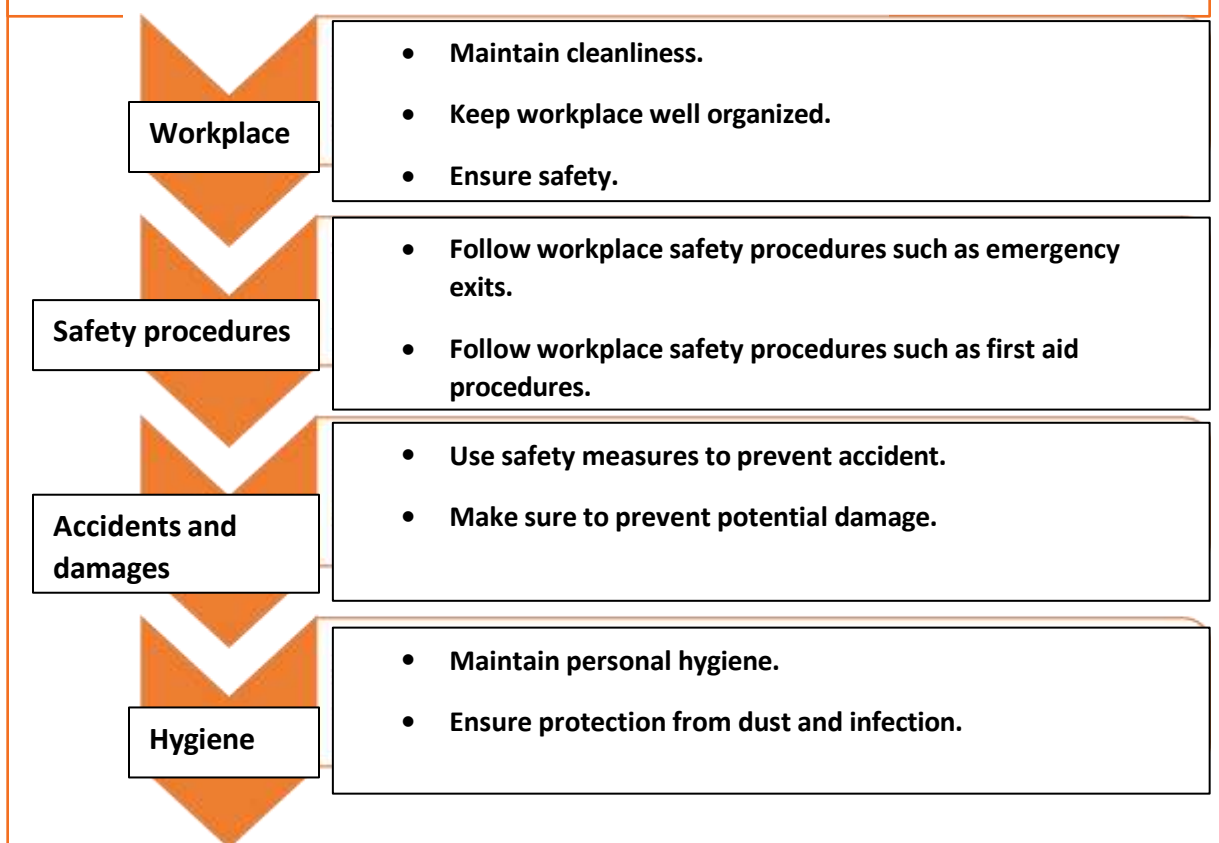
Unit objectives

After reading this unit, you will be able to understand the following points:

1. health and safety standards
2. Describe the safety measures for accident prevention
3. personal health
4. recognition of safety signs
5. Explain the precautions to prevent fire accident
6. identify safety procedures

6.1.1 Occupational health and safety standards

The following safety standards are required in the workplace



Pic.6.1 Occupational health and safety standards

6.1.2 Safety measures for accident prevention

Accidents can be caused by weather conditions, use of machinery, equipment, or vehicles. In fact, non-observance of safety rules or lack of knowledge can also cause injuries. The following measures are necessary for accident prevention:

- **Handle equipment with utmost care and safety.**

- **Equipment should always be usable.**

- **Harmful substances and chemicals should be used carefully.**

- **Such material should never be left unattended in the open.**

- **Personal Protective Equipment (PPE)**

- **PPE must always be in working condition and available at all times.**
- **Must have the necessary information to operate the PPE.**
- **Demonstrate use of safe and essential practices for personal**

6.1.3 Personal hygiene

Personal hygiene should also be taken care of at the workplace like washing hands etc. If the working person is healthy then there are many benefits such as:

- Being able to help co-workers
- Taking few sick leave



Pic.6.2. personal hygiene and cleanliness

6.1.4

Personal Protective Equipment

Personal protective equipment must be worn at all times when performing dislodging activities. A list of required PPE is illustrated below:



Pic.6.3 Personal Protective Equipment ¼PPE½

Item description	Equipment Name	Pictures of equipment
1	Safety helmet or head protection: The worker should use safety helmet while working at the site. Safety helmets help to avoid any head injury.	
2	Safety Goggles: Safety goggles should be worn to protect your eyes.	

3	Nasal Mask: Nasal mask will help protect against bacteria and virus etc.	
4	PVC rubber hand gloves: Gloves will help protect against direct contact with mud.	
5	Acid /base proof suit: Acid /Base proof suit provides protection from direct contact with mud.	

Item description	Equipment Name	Pictures of equipment
6	Gum Boots: The worker should wear gum boots while performing the activities.	
7	Safety Light / Torch: Safety torch can be used properly for inspection.	
8	First Aid Box: First aid box can be used in case of any injury.	

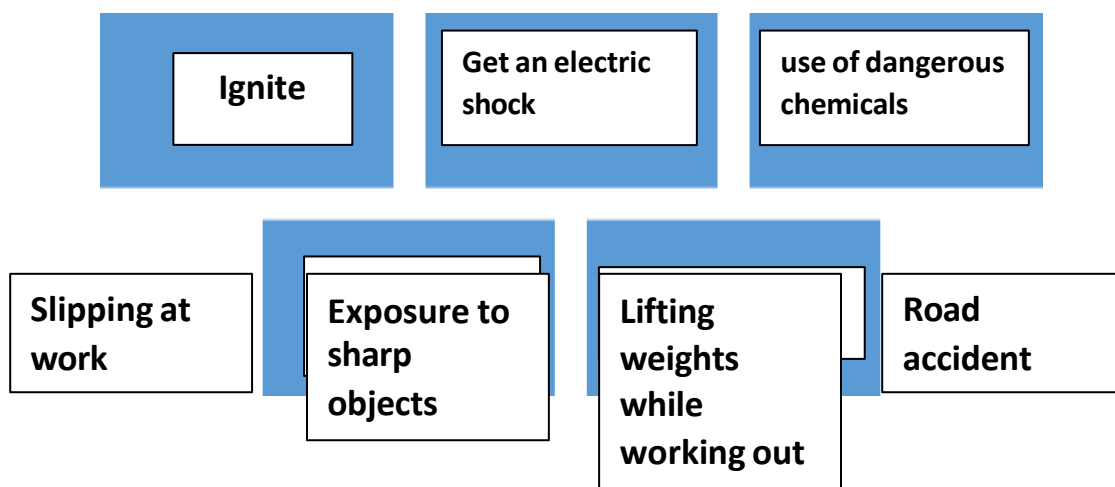
9	Fire Extinguishers: Fire extinguishers can be used in case of fire on site.	
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Pic.6.4 Use of personal protective equipment

6.1.5 Potential hazards

In a workplace, an employee may be exposed to a number of hazards. These hazards can be life-threatening to him or her and to other members.

Some of these hazards are mentioned below:



Pic.6.5 Potential hazards

6.1.5 Point out hazards and use safety signs

Dialog	Communicate workplace hazards and worker-related emergencies
--------	--

The picture below shows the safety signs used in the workplace

- Hot surface



- High voltage



- Inflammable: materials that catch fire



Hazard warning



Biological hazard



Hazardous area



4.6. Safety signs

6.1.6 Emergency contact or communication in danger

serial number	Agency Name	contact number	pictures or sign boards
1	Police: In an emergency, the police will help enforce the law and protect the public.	100	
2	Ambulance: In case of any emergency, the ambulance will help in taking the injured people to the hospital.	102	
3	Fire: In an emergency, the fire brigade will help rescue people.	101	
4	Disaster Management: Disaster Management organization helps people in the event of any natural or man-made disaster.	108	
5	Urban Local Body: ULB is a local government which helps its citizens in case of emergency.	ULB helpline no.	

Table 6.7: Emergency Contact

6.1.7 Fire extinguisher and its types

Fire extinguishers are used to extinguish or control small fires in emergency situations. It is not for use on uncontrolled fires. Fire extinguishers are classified into 4 classes: A, B, C, and D. Each class is effective on a different type of fire. This device and its types are shown in the picture below.



CLASS A



CLASS B



CLASS C



CLASS D

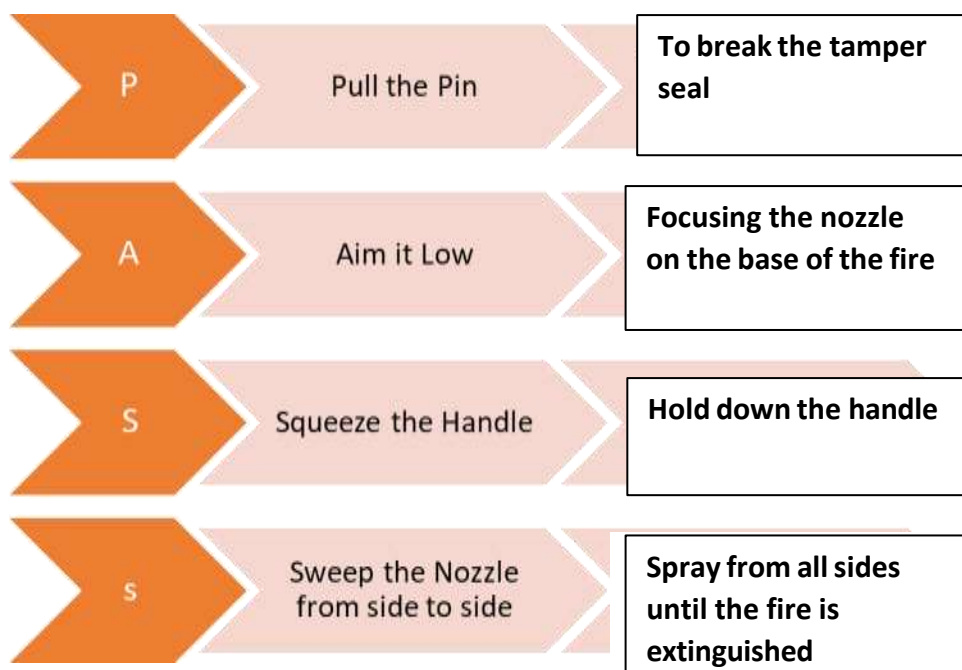
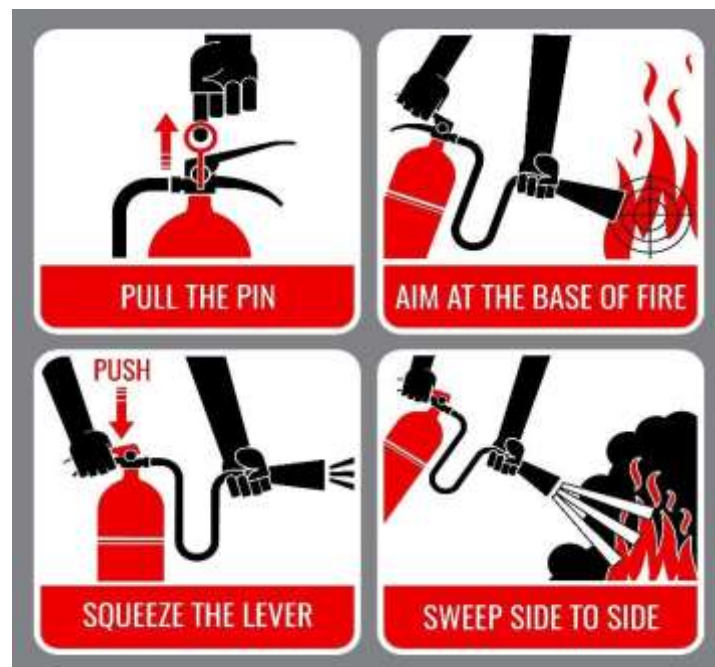


CLASS E



Pic.6.8: Types of fire extinguishers

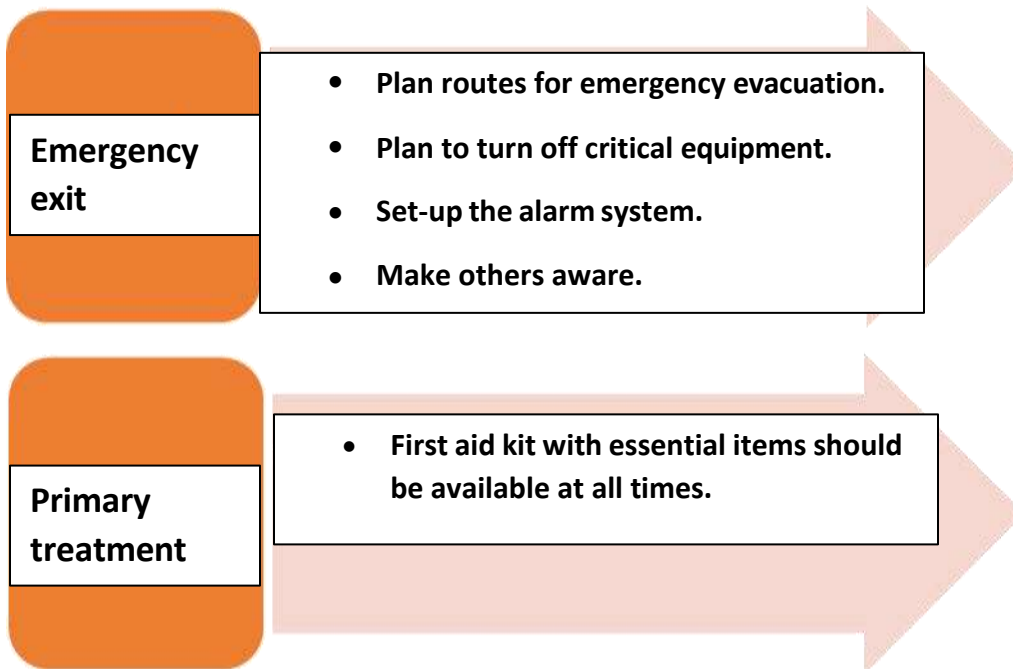
Multipurpose fire extinguishers can be used on different types of fires, such as A.B, B.C or A.B.C. Fire extinguishers are generally very easy to use in the event of a fire. Most operate using P.A.S.S technology. The meaning of P.A.S.S technique is explained below:



Pic.6.9: Operating procedure

6.1.8 Safety procedures

The safety procedure required to avoid any mishap is given below:



Your colleague got hurt he is bleeding profusely from his hand, help him and give him first aid.

Material:

- Cotton
- Clean water
- dettol
- Bandage
- Ointment

Suggestion:

- a. Wash the wound with clean water.
- b. Soak cotton in Dettol and squeeze it. Press the wound with the Dettol cotton and raise your hand slightly.
- c. Press the cotton and tie a bandage on it.
- d. Get medical help as soon as possible if the flow stops or slows down

First aid is the first care given to an injured person before medical help is received. Its objectives:

First aid at the scene of an accident

Prevent injury progression

Quick relief to injured person

Transport the injured person to the nearest hospital

Serial no.	First aid box item name	Pictures
1	First Aid Chart: The adjacent chart details the use of a first aid box in an emergency.	
2	Medical Ointment: Elastic bandages are used without dressing the injured area. It helps in reducing swelling etc.	
3	Medical gauge: small medium and large sterile gauge bandage	
4	Bandages: These bandages are used to hold a dressing over a wound.	

5	Safety Pin: Safety pin is used to fix the bandages.	
---	--	--

Serial no.	First aid box item name	Pictures
6	Disposable Sterile Gloves: These gloves are used in medical exams and prevent cross infection between patients and caregivers.	
7	Scissors: Scissors are used for cutting bandages, gauze and injured area etc.	
8	Alcohol.free cleaning wipe: Wipes are used for cleaning wounds.	
9	Sticky Tape: Sticky tape is used to keep the bandages in place.	
10	Thermometer: Thermometer is used to measure body temperature.	
11	Cream: (Spray antiseptic cream for relief of skin rash, insect bites and sting.	

Serial no.	First aid box item name	Pictures
12	Medicines: (Pain relievers cough antihistamines etc.)	
13	Distilled water for cleaning the wound.	
14	Eye washes and eye baths: This can be used to clean the eyes or soothe irritation.	

Table 6.10: components of first aid box

6.1.9 Immunization Health and Safety

The working person should be protected against Tetanus, Hepatitis A & B, and Diphtheria etc. as they work in very dangerous conditions.



Pic.6.11 Vaccinations to be administered

EXERCISE

Identify and name the safety devices given below.











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2- What are the things to be taken care of under personal hygiene?

- wear clean clothes
- keep your tools and equipment clean
- maintain cleanliness

3- Match the pictures given below with the signs they represent.

	Inflammable material
	Hazard warning
	High voltage
	Danger zone

Experiments

Experiment 1:

Record and document cost, weight and other agricultural residue conditions when stock is collected.

Device:

- pen / laptop/ pc
- office letter form data record document

Security Solution:

- Weight of agri residue w.r.t. Collect status and cost data.
- Enter the details in the official document for the purpose:

Experiment 2:

Use personal protective equipment when cutting agri.residue stock.

Device:

- Glasses
- dust mask
- safety shoes
- safety gloves

Security Solution:

- wear glasses for eye protection
- Wear dust masks for protection against respiratory hazards such as insecticide gases, etc.
- Wear safe shoes to protect your feet from possible wounds.
- Wear protective gloves to protect hands from sharp objects, abrasive surfaces, etc.

Scan the QR code or click on the link to watch the related video

<https://www.youtube.com/watch?v=vWlBgmE6oT8>



<https://www.youtube.com/watch?v=ykmUQc2tpdM>



<https://www.youtube.com/watch?v=L1XS5B5D70U>







7. Employability Skills

It is recommended that all trainings include the appropriate Employability skills Module.

Content for the same can be accessed

[Employability skills workbook](#)





8. Annexure



Module No.	Unit No.	Topic Name	Page no.	URL	QR Code(s)	
1	Unit 1.1	Farm waste and its types	22	https://www.youtube.com/watch?v=Gpo2E.0dDXU		
				https://www.youtube.com/watch?v=2YLPB52zVX8		
	Unit 1.2	Role of Agricultural residue aggregator		https://www.youtube.com/watch?v=um0OIUEX65s		
				https://www.youtube.com/watch?v=WBdVkWxPPFQ		
4	Unit 4.1	Baling and packing agricultural residues	66	https://www.youtube.com/watch?v=yIkYjJsWhek		
				https://www.youtube.com/watch?v=DSP0xk7Lpls&t=120s		

	Unit 4.2	storing agricultural residues		https://www.youtube.com/watch?v=75Rh1Q4Sk6A	
6	Unit 6.1	maintaining health and safety at work	102	https://www.youtube.com/watch?v=vWlBgmE6oT8	
				https://www.youtube.com/watch?v=ykmUQc2tpdM	
				https://www.youtube.com/watch?v=L1XS5B5D70U	



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